

BPSC TRE 4.0 – Computer Teacher

Tutorial + Practice Questions | English + हिंदी

Based on the topics asked in BPSC TRE previous year

Computer Science papers (see the **papers** folder)

How to use / कैसे पढ़ें: Read each chapter, then solve the practice MCQs at the end and check the answers. Every BPSC question has options (A)–(E): (D) = "More than one of the above", (E) = "None of the above" — always check all options before choosing.

हर अध्याय पढ़ें, फिर अंत के MCQ हल करें। BPSC में विकल्प (D) "उपर्युक्त में से एक से अधिक" और (E) "उपर्युक्त में से कोई नहीं" होते हैं — सभी विकल्प जाँचकर ही उत्तर चुनें।

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1. Number Systems & Digital Logic / संख्या पद्धति और डिजिटल लॉजिक

Number systems / संख्या पद्धतियाँ

System	Base	Digits
Binary / द्वि-आधारी	2	0, 1
Octal / अष्ट-आधारी	8	0–7
Decimal / दशमलव	10	0–9
Hexadecimal / षोडश-आधारी	16	0–9, A–F (A=10 ... F=15)

- Decimal → binary: divide by 2 repeatedly, read remainders bottom to top. दशमलव → बाइनरी: बार-बार 2 से भाग दें, शेषफल नीचे से ऊपर पढ़ें।
- Binary → octal: group 3 bits; binary → hex: group 4 bits (from the right). बाइनरी → ऑक्टल: 3 बिट के समूह; बाइनरी → हेक्स: 4 बिट के समूह।
- Fraction: $1011.1101_2 = 8+2+1 + 0.5+0.25+0.0625 = 11.8125$
- 1's complement = flip all bits; 2's complement = 1's complement + 1. Negative numbers are stored in 2's complement. ऋणात्मक संख्याएँ 2's complement में संग्रहित होती हैं।
- Range of n-bit signed (2's complement): -2^{n-1} to $2^{n-1}-1$.
- Booth's algorithm = multiplication of signed binary numbers. बूथ एल्गोरिदम = बाइनरी गुणा।
- Error detection: parity bit, checksum, CRC. Error correction: Hamming code.
- Codes: ASCII (7-bit, 128 chars), Extended ASCII (8-bit, 256), Unicode (supports Hindi and all languages), BCD, Gray code.

Logic gates / लॉजिक गेट

Gate	Expression	Output is 1 when...
AND	$A \cdot B$	both inputs 1
OR	$A + B$	any input 1
NOT	A'	input is 0
NAND	$(A \cdot B)'$	any input 0 — universal gate
NOR	$(A + B)'$	both inputs 0 — universal gate
XOR	$A \oplus B$	inputs are different (odd number of 1s)
XNOR	$(A \oplus B)'$	inputs are same

- NAND and NOR are universal gates — any circuit can be built from them alone. **NAND और NOR यूनिवर्सल गेट हैं।**
- De Morgan: $(A \cdot B)' = A' + B'$ and $(A + B)' = A' \cdot B'$. Used to simplify Boolean expressions. **बूलियन व्यंजकों को सरल करने में उपयोगी।**
- Don't-care condition: input combination whose value does not affect the output (used in K-map to simplify).
- Half adder: Sum = $A \oplus B$, Carry = $A \cdot B$. Full adder adds 3 bits (A, B, Cin).
- Multiplexer: 2^n -to-1 needs n select lines (8-to-1 $\rightarrow$ 3). Decoder: n inputs $\rightarrow 2^n$ outputs.
- Flip-flops: SR, JK, D, T (store 1 bit). Combinational circuit: output depends only on current input; sequential: also on past state (has memory).
- PLA = Programmable Logic Array. ROM types: MROM, PROM, EPROM, EEPROM.

2. Computer Fundamentals & Memory / कंप्यूटर के मूल सिद्धांत और मेमोरी

- Generations: 1st vacuum tube, 2nd transistor, 3rd IC, 4th microprocessor (VLSI), 5th AI. पहली पीढ़ी वैक्यूम ट्यूब, दूसरी ट्रांजिस्टर, तीसरी IC, चौथी माइक्रोप्रोसेसर, पाँचवीं AI।
- CPU = ALU + Control Unit + Registers. Von Neumann architecture: program and data stored in the same memory.
- Memory hierarchy (fast → slow, small → big): Registers → Cache → RAM → Secondary storage (HDD/SSD). रजिस्टर → कैश → RAM → हार्ड डिस्क।
- RAM: volatile (data lost when power off). ROM: non-volatile. SRAM (cache, faster, costly), DRAM (main memory, needs refresh).
- 1 KB = 1024 B, 1 MB = 1024 KB, 1 GB = 1024 MB, 1 TB = 1024 GB. 1 byte = 8 bits; 4 bits = nibble.
- Cache mapping: direct, associative, set-associative. TLB is an associative cache for page table entries.
- Input devices: keyboard, mouse, scanner, OMR, OCR, MICR, barcode reader. Output: monitor, printer, plotter, speaker.
- Software: System software (OS, compiler, assembler, interpreter, utilities) and application software. Compiler translates the whole program; interpreter line by line; assembler converts assembly to machine code.
- Booting: BIOS/POST → bootloader → OS loaded into RAM. Cold boot = power on; warm boot = restart.
- Instruction cycle: Fetch → Decode → Execute.

3. C Programming / C प्रोग्रामिंग

Basics

- C was developed by Dennis Ritchie at Bell Labs (1972). Case-sensitive; execution starts at `main()`. C भाषा डेनिस रिची ने बनाई; क्रियान्वयन `main()` से शुरू होता है।
- Data types: `int` (2/4 B), `char` (1 B), `float` (4 B), `double` (8 B). Format specifiers: `%d`, `%f`, `%c`, `%s`, `%lf`, `%x`.
- Operators: arithmetic (+ - * / %), relational, logical (&& || !), bitwise (& | ^ ~ << >>), assignment, ++/--, ternary (?:), `sizeof`.
- Integer division: $7/2 = 3$; $7\%2 = 1$; $7/2.0 = 3.5$.
- Operator precedence (high → low): `() []` → unary (`++ -- ! ~`) → `* / %` → `+ -` → shift → relational → `== !=` → `&` → `^` → `|` → `&&` → `||` → `?:` → `=`
- Storage classes: `auto`, `register`, `static` (keeps value between calls), `extern`.
- Loops: `for`, `while` (entry-controlled), `do-while` (exit-controlled, runs at least once). `break` exits loop; `continue` skips to next iteration.

Arrays, pointers, functions, strings

- Array index starts at 0; array name is the address of the first element. `int a[5] → a[0]...a[4]`.
- Pointer stores an address: `int *p = &x`; `*p` gives value, `p` gives address. `p+1` moves by `sizeof(type)` bytes. पॉइंटर पता संग्रहित करता है; `*p` मान देता है।
- Call by value (copy) vs call by reference (pass pointer, original changes).
- String = `char` array ending with `'\0'`. Functions: `strlen`, `strcpy`, `strcat`, `strcmp` (`string.h`).

- malloc(size) allocates memory (uninitialised), calloc(n,size) (zero-initialised), realloc resizes, free releases. Memory leak = allocated but never freed.
- Recursion: a function calling itself; needs a base case. Factorial: $f(n) = n \times f(n-1)$, $f(0)=1$.
- struct groups different types; union shares one memory for all members (size = largest member).

```
#include <stdio.h>
int main() {
    int a[] = {1, 2, 3}, *p = a;
    printf("%d %d", *p, *(p + 2));    // output: 1 3
    return 0;
}
```

Tip: $5/2 \rightarrow 2$, $-5/2 \rightarrow -2$ (truncates toward zero); $x++$ uses then increments, $++x$ increments then uses.

सुझाव: $x++$ पहले उपयोग, फिर बढ़ोतरी; $++x$ पहले बढ़ोतरी, फिर उपयोग।

4. Data Structures / डेटा स्ट्रक्चर

Structure	Key points / मुख्य बिंदु
Array	Fixed size, contiguous memory, $O(1)$ access by index.
Linked list	Nodes with data + pointer; dynamic size; insertion/deletion $O(1)$ at known position; access $O(n)$. Types: singly, doubly, circular.
Stack	LIFO (Last In First Out); push, pop, peek. Uses: function calls, undo, expression evaluation, DFS.
Queue	FIFO; enqueue, dequeue. Types: circular, deque, priority queue. Uses: scheduling, BFS.
Tree	Non-linear hierarchical structure. Root, parent, child, leaf, height.
Graph	Vertices + edges; non-linear. Stored as adjacency matrix or adjacency list.
Hash table	Key $\rightarrow$ index by hash function; average $O(1)$ search, insert, delete. Collisions: chaining, open addressing.

- Linear: array, list, stack, queue. Non-linear: tree, graph.
रैखिक: ऐरे, लिस्ट, स्टैक, क्यू; गैर-रैखिक: ट्री, ग्राफ।
- Binary tree: max 2 children per node. Max nodes at level $i = 2^i$; max nodes in height $h = 2^{h+1} - 1$.
- BST: left < root < right. Inorder traversal of BST gives sorted order. Number of distinct BSTs with n keys = Catalan number: $n=3 \rightarrow 5$, $n=4 \rightarrow 14$.
- Traversals: Inorder (L-Root-R), Preorder (Root-L-R), Postorder (L-R-Root). BFS uses a queue (level order); DFS uses a stack.
- Balanced tree (AVL): height difference of the two subtrees of any node ≤ 1 . Heap: complete binary tree; max-heap parent $\geq$ children.
- Spanning tree of a graph with N vertices has $N-1$ edges. Complete graph has $N(N-1)/2$ edges.

- Postfix: operands first, then operator. Evaluate with stack. Example: $10\ 5\ + \rightarrow 15$; $60\ 6\ /\rightarrow 10$; then $15\ 10\ *\rightarrow 150$.
- Infix to postfix: $A+B*C \rightarrow ABC*+ ;$ prefix: $+A*BC$.

5. Algorithms & Complexity / एल्गोरिदम और जटिलता

Algorithm	Best	Average	Worst	Note
Linear search	1	n	n	works on unsorted data
Binary search	1	$\log n$	$\log n$	needs sorted data
Bubble / Selection / Insertion sort	n (bubble, insertion)	n^2	n^2	simple, in-place
Merge sort	$n \log n$	$n \log n$	$n \log n$	Divide & Conquer, needs extra space
Quick sort	$n \log n$	$n \log n$	n^2	Divide & Conquer, pivot based
Heap sort	$n \log n$	$n \log n$	$n \log n$	uses heap

- Big-O = upper bound (worst case); Omega (Ω) = lower bound; Theta (Θ) = tight bound. बिग-ओ ऊपरी सीमा दर्शाता है।
- Order: $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n) < O(n!)$.
- Divide and Conquer: Divide $\rightarrow$ Conquer $\rightarrow$ Combine (merge sort, quick sort, binary search).

- Greedy: locally best choice (Prim's, Kruskal's, Dijkstra, Huffman). Dynamic Programming: stores sub-problem results (Fibonacci, knapsack, LCS).
- Graph traversal: BFS and DFS. Shortest path: Dijkstra, Bellman-Ford, Floyd-Warshall. Minimum spanning tree: Prim, Kruskal.
- Binary search on n elements needs at most $\lfloor \log_2 n \rfloor + 1$ comparisons.

6. Operating System / ऑपरेटिंग सिस्टम

- OS = interface between user and hardware; manages process, memory, file, I/O. Types: batch, multiprogramming, time-sharing, real-time, distributed, network, embedded, mobile. OS उपयोगकर्ता और हार्डवेयर के बीच इंटरफेस है।
- Process states: New $\rightarrow$ Ready $\rightarrow$ Running $\rightarrow$ Waiting/Blocked $\rightarrow$ Terminated. PCB (Process Control Block) stores process information.
- Process vs thread: thread is a lightweight process sharing code, data and files of its process but with its own stack and registers.

CPU scheduling / सीपीयू शेड्यूलिंग

- FCFS (non-preemptive, convoy effect), SJF (minimum average waiting time; may starve), Priority (starvation $\rightarrow$ solved by aging), Round Robin (time quantum; preemptive; good for time-sharing).
- Turnaround time = Completion - Arrival; Waiting time = Turnaround - Burst.

Deadlock / डेडलॉक

- Four necessary conditions (all must hold): Mutual exclusion, Hold and wait, No preemption, Circular wait.
- Handling: prevention, avoidance (Banker's algorithm), detection and recovery, ignore (ostrich).
- Semaphore: integer variable with wait(P) and signal(V); binary semaphore (0/1) and counting semaphore.
Critical section problem: mutual exclusion, progress, bounded waiting.

Memory management / मेमोरी प्रबंधन

- Paging: physical memory divided into fixed-size **frames**, logical memory into **pages** of same size; no external fragmentation (internal fragmentation possible). पेजिंग: निश्चित आकार के पेज और फ्रेम।
- Segmentation: variable-size logical segments (code, data, stack); segment table holds base and limit.
- Virtual memory: demand paging; page fault when needed page is not in RAM. Thrashing = excessive paging, CPU spends more time swapping than executing.
- Page replacement: FIFO (Belady's anomaly), LRU, Optimal.
- Page offset bits = $\log_2(\text{page size})$. Page size 4 KB = 2^{12} → 12 offset bits.
- Allocation: first fit, best fit, worst fit. External fragmentation (free space scattered) vs internal (wasted inside allocated block).

File systems / फाइल सिस्टम

- Windows: FAT32, NTFS, exFAT. Linux: ext3/ext4. macOS: HFS+/APFS. File operations: create, open, read, write, delete, rename, seek.

- Common Linux commands: ls, cd, pwd, mkdir, rm, cp, mv, cat, chmod, grep, ps, kill.

7. DBMS & SQL / डेटाबेस और SQL

- DBMS manages a database; examples: MySQL, Oracle, PostgreSQL, SQL Server, MS Access. Advantages: less redundancy, data integrity, security, sharing, backup.
- Data models: hierarchical, network, relational (most used), object-oriented. ER model: entity, attribute, relationship; cardinality 1:1, 1:N, M:N.
- Keys: Super key, Candidate key, **Primary key** (unique, not null), Alternate key, Foreign key (refers to primary key of another table), Composite key.
- Degree = number of columns (attributes); Cardinality = number of rows (tuples). डिग्री = कॉलम की संख्या; कार्डिनैलिटी = पंक्तियों की संख्या।

Normalization / सामान्यीकरण

Form	Rule
1NF	Every attribute atomic (single valued), no repeating groups
2NF	1NF + no partial dependency on part of the candidate key
3NF	2NF + no transitive dependency
BCNF	For every dependency $X \rightarrow Y$, X is a super key

SQL

Type	Commands
DDL	CREATE, ALTER, DROP, TRUNCATE, RENAME
DML	INSERT, UPDATE, DELETE, SELECT
DCL	GRANT, REVOKE
TCL	COMMIT, ROLLBACK, SAVEPOINT

```
CREATE TABLE student (roll INT PRIMARY KEY, name  
VARCHAR(30), marks INT);  
INSERT INTO student VALUES (1, 'Rahul', 85);  
SELECT name, marks FROM student WHERE marks > 60 ORDER BY  
marks DESC;  
SELECT COUNT(*), AVG(marks), MAX(marks) FROM student;  
SELECT dept, COUNT(*) FROM emp GROUP BY dept HAVING  
COUNT(*) > 2;  
UPDATE student SET marks = 90 WHERE roll = 1;  
DELETE FROM student WHERE roll = 1;
```

- WHERE filters rows (before grouping); HAVING filters groups (after GROUP BY). Aggregate functions: COUNT, SUM, AVG, MIN, MAX.
- Joins: INNER, LEFT, RIGHT, FULL OUTER, CROSS. DELETE removes rows (can rollback); TRUNCATE removes all rows (no rollback in most DBs); DROP removes the table itself.
- ACID: Atomicity, Consistency, Isolation, Durability.

8. Computer Networks / कंप्यूटर नेटवर्क

- Types: PAN, LAN (building), MAN (city), WAN (country; Internet is the largest WAN).
- Topologies: Bus, Star (most common; central hub/switch), Ring, Mesh ($n(n-1)/2$ links), Tree, Hybrid.
- Devices: Hub (broadcasts), Switch (uses MAC address), Router (uses IP address, connects networks), Bridge, Gateway, Modem, Repeater.

OSI model (7 layers) / OSI मॉडल

#	Layer	Role
7	Application	HTTP, FTP, SMTP, DNS
6	Presentation	encryption, compression, translation
5	Session	session / dialogue control
4	Transport	TCP, UDP; end-to-end delivery
3	Network	IP, routing, logical address
2	Data link	MAC address, framing, error detection
1	Physical	bits, cables, signals

Memory trick / याद रखने की ट्रिक: **P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way (1→7).

- TCP/IP model has 4 layers: Application, Transport, Internet, Network access.
- TCP: reliable, connection-oriented. UDP: unreliable, connectionless, faster (video streaming).
- IPv4 = 32-bit (e.g., 192.168.1.1), IPv6 = 128-bit. MAC address = 48-bit physical address.
- IP classes: A (1–126), B (128–191), C (192–223), D (multicast 224–239), E (reserved).
- Ports: HTTP 80, HTTPS 443, FTP 21, SSH 22, SMTP 25, DNS 53, POP3 110, Telnet 23.
- DNS converts domain names to IP address. DHCP assigns IP automatically. ARP maps IP → MAC.
- Transmission modes: simplex (one way), half duplex (both ways, one at a time), full duplex (both ways at once).
- Media: twisted pair, coaxial, optical fibre (fastest, no EMI), wireless (Wi-Fi, Bluetooth, microwave).

9. Internet, Web & Security / इंटरनेट, वेब और सुरक्षा

- WWW invented by Tim Berners-Lee (1989–91).
Browser examples: Chrome, Firefox, Edge. URL = protocol + domain + path.
- HTML = markup language for web pages; tags: <h1>–<h6>, <p>, <a href>, , <table>, /, <form>. CSS styles the page; JavaScript adds behaviour. **HTML वेब पेज की संरचना बनाती है, CSS सजावट, JavaScript व्यवहार।**
- Email: SMTP sends; POP3/IMAP receive. To, Cc, Bcc (hidden recipients).
- Security threats: virus, worm (self-replicating), trojan (disguised), ransomware, spyware, phishing (fake emails/sites), spoofing, DoS/DDoS, man-in-the-middle.
- Protection: antivirus, firewall, strong passwords, two-factor authentication, encryption (symmetric: one key – AES/DES; asymmetric: public + private key – RSA), digital signature, HTTPS/SSL/TLS.
- Cyber crime law in India: IT Act 2000. E-commerce, UPI, digital India, e-governance, cloud computing (IaaS, PaaS, SaaS), IoT, AI, Big Data, blockchain – basic definitions are often asked.
- MS Office shortcuts: Ctrl+C copy, Ctrl+X cut, Ctrl+V paste, Ctrl+Z undo, Ctrl+Y redo, Ctrl+S save, Ctrl+P print, Ctrl+A select all, Ctrl+B/I/U bold/italic/underline, F12 Save As (Word). Excel: formulas start with "=", e.g. =SUM(A1:A5), =AVERAGE(), =IF(), =VLOOKUP(); cell reference A1; extension .xlsx; Word .docx; PowerPoint .pptx.

10. OOP, C++ & Python / ओओपी, C++ और पाइथन

- OOP pillars: **Encapsulation** (data hiding in class), **Abstraction**, **Inheritance** (reuse; single, multiple, multilevel, hierarchical, hybrid), **Polymorphism** (compile-time: overloading; run-time: overriding, virtual function).
- C++ was created by Bjarne Stroustrup. Class members default to private (struct → public). Constructor has the same name as the class, no return type; destructor uses ~ClassName. Access specifiers: private, protected, public.
- Java: platform independent via bytecode and JVM; no pointers; garbage collection. Developed by James Gosling at Sun.

Python basics

- Interpreted, dynamically typed, indentation defines blocks. Created by Guido van Rossum. Comments start with #.
- Data types: int, float, str, bool, list [] (mutable), tuple () (immutable), set { } (unique), dict {key: value}.
- Loops: for i in range(5) → 0..4; while; break/continue. Functions: def f(x): return x*2.

```
nums = [1, 2, 3, 4]
print(nums[-1])           # 4
print(nums[1:3])          # [2, 3]
print([x*x for x in nums]) # [1, 4, 9, 16]
d = {"a": 1}; d["b"] = 2
print(len(d))             # 2
print(10 // 3, 10 % 3, 2 ** 3) # 3 1 8
```

11. Practice MCQs / अभ्यास प्रश्न

Answers are given at the end of this section. / उत्तर इस भाग के अंत में दिए गए हैं।

1. Which gate is called a universal gate? / कौन-सा गेट यूनिवर्सल गेट कहलाता है?

(A) AND (B) OR (C) NAND (D) XOR

2. Binary of decimal 25 / दशमलव 25 का बाइनरी:

(A) 11001 (B) 10011 (C) 11010 (D) 10101

3. Hexadecimal digit F equals decimal / हेक्साडेसिमल F का दशमलव मान:

(A) 14 (B) 15 (C) 16 (D) 13

4. How many select lines are needed for a 16-to-1 multiplexer? / 16-to-1 मल्टीप्लेक्सर में कितनी सेलेक्ट लाइनें?

(A) 2 (B) 3 (C) 4 (D) 8

5. Which memory is volatile? / कौन-सी मेमोरी वोलाटाइल है?

(A) ROM (B) RAM (C) PROM (D) EPROM

6. Fastest memory in a computer / कंप्यूटर की सबसे तेज मेमोरी:

(A) RAM (B) Cache (C) Register (D) Hard disk

7. Output of $7/2$ in C (int) / C में $7/2$ का आउटपुट:

(A) 3.5 (B) 3 (C) 4 (D) 2

8. A C string ends with / C स्ट्रिंग किससे समाप्त होती है:

(A) '\n' (B) '\0' (C) ' ' (D) EOF

9. Which loop executes at least once? / कौन-सा लूप कम से कम एक बार चलता है?

(A) for (B) while (C) do-while (D) none

- 10.** Which function allocates zero-initialised memory? / शून्य से प्रारंभ की गई मेमोरी कौन-सा फ़ंक्शन देता है?
(A) malloc (B) calloc (C) realloc (D) free
- 11.** Stack follows / स्टैक किस सिद्धांत पर चलता है:
(A) FIFO (B) LIFO (C) Random (D) Priority
- 12.** Inorder traversal of a BST gives / BST का इनऑर्डर ट्रेवर्सल देता है:
(A) Reverse order (B) Sorted order (C) Level order (D) Random order
- 13.** Postfix value of $2\ 3\ 4\ * +$ / पोस्टफिक्स $2\ 3\ 4\ * +$ का मान:
(A) 14 (B) 20 (C) 9 (D) 24
- 14.** Number of edges in a spanning tree of 10 vertices / 10 शीर्षों वाले ग्राफ के स्पैनिंग ट्री में किनारे:
(A) 10 (B) 9 (C) 11 (D) 45
- 15.** Data structure used in BFS / BFS में प्रयुक्त डेटा स्ट्रक्चर:
(A) Stack (B) Queue (C) Tree (D) Array
- 16.** Worst case time of binary search / बाइनरी सर्च का सबसे खराब समय:
(A) $O(n)$ (B) $O(\log n)$ (C) $O(n \log n)$ (D) $O(1)$
- 17.** Merge sort is based on / मर्ज सॉर्ट आधारित है:
(A) Greedy (B) Divide and Conquer (C) Dynamic programming (D) Backtracking
- 18.** Worst case of quick sort / क्विक सॉर्ट का सबसे खराब समय:
(A) $O(n \log n)$ (B) $O(n)$ (C) $O(n^2)$ (D) $O(\log n)$
- 19.** Which is NOT a necessary condition for deadlock? / डेडलॉक की आवश्यक शर्त कौन-सी नहीं है?
(A) Mutual exclusion (B) Hold and wait (C) Preemption (D) Circular wait

20. Banker's algorithm is used for / बैंकर्स एल्गोरिदम का उपयोग:

(A) Deadlock avoidance (B) Paging (C) Scheduling (D) Sorting

21. Excessive paging activity is called / अत्यधिक पेजिंग को कहते हैं:

(A) Spooling (B) Thrashing (C) Fragmentation (D) Caching

22. Page size 8 KB → number of offset bits / पेज साइज 8 KB → ऑफसेट बिट्स:

(A) 12 (B) 13 (C) 14 (D) 10

23. Scheduling algorithm with time quantum / टाइम क्वांटम वाला शेड्यूलिंग एल्गोरिदम:

(A) FCFS (B) SJF (C) Round Robin (D) Priority

24. Which key uniquely identifies a row and cannot be NULL? / कौन-सी की पंक्ति को अद्वितीय पहचानती है और NULL नहीं हो सकती?

(A) Foreign key (B) Primary key (C) Alternate key (D) Super key

25. Which SQL clause filters groups? / कौन-सा SQL क्लॉज़ समूहों को फ़िल्टर करता है?

(A) WHERE (B) HAVING (C) ORDER BY (D) LIMIT

26. Which is a DDL command? / DDL कमांड कौन-सा है?

(A) INSERT (B) UPDATE (C) ALTER (D) SELECT

27. 3NF removes / 3NF हटाता है:

(A) Partial dependency (B) Transitive dependency (C) Multivalued attribute (D) Foreign key

28. ACID: the letter 'I' stands for / ACID में 'I':

(A) Integrity (B) Isolation (C) Indexing (D) Inheritance

29. Which layer of OSI handles routing? / OSI की कौन-सी परत रूटिंग करती है?

(A) Data link (B) Transport (C) Network (D) Session

30. Default port of HTTPS / HTTPS का डिफ़ॉल्ट पोर्ट:

(A) 80 (B) 21 (C) 443 (D) 25

31. IPv6 address size / IPv6 पते का आकार:

(A) 32 bit (B) 64 bit (C) 128 bit (D) 256 bit

32. Which device works with MAC addresses? / कौन-सा उपकरण MAC पते से काम करता है?

(A) Hub (B) Switch (C) Repeater (D) Modem

33. Protocol used for sending email / ईमेल भेजने वाला प्रोटोकॉल:

(A) POP3 (B) IMAP (C) SMTP (D) FTP

34. Self-replicating malware that spreads over a network / नेटवर्क पर स्वयं फैलने वाला मैलवेयर:

(A) Worm (B) Trojan (C) Spyware (D) Adware

35. RSA is a / RSA है:

(A) Symmetric key algorithm (B) Asymmetric key algorithm (C) Hash function (D) Compression method

36. Which is not an OOP principle? / कौन-सा OOP सिद्धांत नहीं है?

(A) Encapsulation (B) Inheritance (C) Compilation (D) Polymorphism

37. Python output of `print(10 // 3)` / पाइथन में `print(10 // 3)`:

(A) 3.33 (B) 3 (C) 4 (D) 1

38. Which Python type is immutable? / कौन-सा पाइथन टाइप अपरिवर्तनीय है?

(A) list (B) dict (C) tuple (D) set

39. Shortcut for Undo in MS Word / MS Word में Undo का शॉर्टकट:

(A) Ctrl+Y (B) Ctrl+Z (C) Ctrl+U (D) Ctrl+X

40. Extension of Excel 2007+ file / एक्सेल 2007+ फाइल का एक्सटेंशन:

(A) .xls (B) .xlsx (C) .docx (D) .pptx

Answers / उत्तर: 1-C, 2-A ($16+8+1=25$), 3-B, 4-C ($2^4=16$), 5-B, 6-C, 7-B, 8-B, 9-C, 10-B, 11-B, 12-B, 13-A ($3 \times 4=12$, $+2=14$), 14-B ($N-1$), 15-B, 16-B, 17-B, 18-C, 19-C (Preemption is allowed $\rightarrow$ "No preemption" is the condition), 20-A, 21-B, 22-B ($8KB=2^{13}$), 23-C, 24-B, 25-B, 26-C, 27-B, 28-B, 29-C, 30-C, 31-C, 32-B, 33-C, 34-A, 35-B, 36-C, 37-B, 38-C, 39-B, 40-B

12. Solved previous-year questions / हल किए गए पिछले वर्ष के प्रश्न

From the BPSC Class 11–12 Computer Science paper (22 July 2024) in your **papers** folder. Question numbers match that paper. Only questions whose answer could be worked out with certainty are listed here — for the rest, use the official answer-key PDFs in the same folder.

ये प्रश्न papers फ़ोल्डर के 22 जुलाई 2024 वाले पेपर से हैं। बाकी प्रश्नों के लिए उसी फ़ोल्डर की Answer Key देखें।

Q.	Question	Answer / Working
76	1010 AND 1100 in binary	(A) 1000 — bit-wise AND
80	Select lines for an 8-to-1 multiplexer	(B) 3 — $2^3 = 8$
81	Decimal 1234 → binary, octal, hex	(A) binary 10011010010, octal 2322, hex 4D2
82	Booth's algorithm is used for	(C) binary multiplication
83	Error detection code in digital logic	(D) more than one — Hamming, CRC and checksum are all used
85	1011.1101 (binary) → decimal	$8+2+1+0.5+0.25+0.0625 = 11.8125 \rightarrow$ none of 11.75 / 12.125 / 13.5 → (E)
86	De Morgan's theorem is used for	(A) simplifying Boolean expressions (it also helps implement with NAND gates, so check the official key for (A) vs (D))
88	Non-linear data structure	Tree/graph — none of Stack, Queue, Array → (E)
92	Balanced binary tree: subtree height difference at most	(B) 1
96	Preorder 15,10,12,11,20,18,16,19 → postorder	Root 15; left subtree {10,12,11}; right {20,18,16,19} → (B) 11,12,10,16,19,18,20,15
97	Distinct BSTs with 4 keys	Catalan $C_4 = 14 \rightarrow$ (C)
99	Big-O notation	(B) upper bound of runtime
101	Merge sort paradigm	(B) Divide and Conquer
102	Edges of spanning tree with N vertices	(B) N - 1
112	Page size 4 KB: page offset bits	4 KB = $2^{12} \rightarrow$ (B) 12 bits
114	Cause of thrashing	(A) excessive paging activity

116	TLB organisation	Typically an associative cache; check the official key (B vs C)
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Study plan / पढ़ाई की योजना

- Week 1: chapters 1–2 + 20 PYQs from the papers folder. सप्ताह 1: अध्याय 1–2।
- Week 2: chapters 3–5 (C, data structures, algorithms) — write and trace small programs by hand.
- Week 3: chapters 6–7 (OS, DBMS) — practise SQL queries on paper.
- Week 4: chapters 8–10, then solve one full previous-year paper in a timed mode (2.5 hours, 150 questions).
- Always check the official syllabus and notification on bpsc.bihar.gov.in. अंतिम पाठ्यक्रम BPSC की वेबसाइट से जाँचें।

13. Video links / वीडियो लिंक (tap to open)

- [TRE 4.0 & STET CS – Shubham Sir](#)
- [Mindmap Gurukul – TRE 4.0 CS](#)
- [Most Important MCQs Part 1](#)
- [Data Structure for TRE 4.0 \(L-5\).](#)
- [121-day strategy](#)
- [BPSC TRE 4.0 Computer Teacher](#)
- [DBMS complete playlist](#)
- [DSA – CodeWithHarry](#)
- [DSA \(C++\).](#)
- [C Programming & Data Structures](#)